

Build-A-Job

Product Requirements Document *AI-Powered Trade Marketplace Platform*

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1. Executive Summary

Build-A-Job is an AI-powered two-sided marketplace that connects homeowners (customers) with verified trade professionals across the United Kingdom. The platform eliminates the friction in the home improvement journey by combining AI-driven design visualisation, instant cost estimation, smart job-matching, and a transparent quoting workflow — all within a single web and mobile application.

This Product Requirements Document (PRD) covers the complete product definition across three distinct application surfaces:

- Customer App (Web + iOS + Android) — the homeowner-facing interface
- Trade Professional App (Web + iOS + Android) — the tradesperson-facing interface
- Admin Dashboard (Web) — the internal operations and platform management console

Vision: To become the default platform that homeowners turn to when they imagine a home project — and that trade professionals turn to for a steady pipeline of qualified, well-briefed leads.

The platform is being built in three phases:

- Phase 1 — MVP & Pilot: driveways and key exterior trades in select UK cities
- Phase 2 — Scale by Trade & Region: broader categories and national rollout
- Phase 3 — Advanced AI & Ecosystem: AR previews, supplier integrations, white-labelling

2. Document Information

2.1 Purpose

This PRD serves as the single source of truth for all product, design, and engineering decisions for the Build-A-Job platform. It defines scope, personas, journeys, functional requirements, non-functional requirements, and acceptance criteria to be used by all stakeholders throughout the project lifecycle.

2.2 Scope

This document covers version 1.0 (MVP) of the Build-A-Job platform with forward-looking notes for Phases 2 and 3 where relevant.

2.3 Document Conventions

Convention	Meaning
MUST	Mandatory requirement — must be delivered in Phase 1 MVP
SHOULD	Highly desirable — included if schedule allows
COULD	Optional enhancement for future phases
P0	Critical — blocker to launch
P1	High priority — required for a quality launch
P2	Medium priority — improves experience post-launch

3. Product Overview

3.1 Problem Statement

Home improvement projects in the UK represent a high-value market, yet the process of finding, briefing, and engaging a trade professional remains deeply inefficient:

- Customers cannot clearly articulate what they want or what a realistic budget looks like.
- Trade professionals waste significant time on unqualified leads, vague briefs, and no-shows.
- No existing platform combines visual job briefing, AI design previews, and smart cost estimation.
- Current players (Checkatrade, MyBuilder, Rated People) focus on listings and reviews — not on helping customers define the job before it reaches a trade.

3.2 Proposed Solution

Build-A-Job solves this by sitting before the traditional lead-generation step. When a customer uploads a photo of their driveway, garden, or room, the platform analyses it, generates multiple design concepts, attaches an instant estimate range, and packages the entire brief into a structured job request. This brief — rich with visuals and budget context — is what reaches the trade professional.

3.3 Key Differentiators

- AI-generated before/after design previews from a single photo

- Instant estimate ranges calibrated to UK materials and labour costs
- Structured job packaging that pre-qualifies leads before they reach trades
- Trade-facing AI tools (alternative design suggestions when quoting) to win more work
- End-to-end platform from imagination through design, quote, hire, and review

3.4 Platform Architecture Overview

Build-A-Job comprises three interconnected applications sharing a common API and data layer:

Application	Primary Users	Core Function
Customer App	Homeowners / General Public	Discover, design, request quotes, review trades
Trade Professional App	Builders, Landscapers, Tradespeople	Manage profile, receive leads, quote, build reputation
Admin Dashboard	Build-A-Job Internal Team	Platform management, moderation, analytics, configuration

4. User Personas

The following personas represent the core user archetypes across the Build-A-Job platform. Each feature and user story should be validated against these personas.

4.1 Customer Personas

Persona 1 — The Inspired Homeowner

Claire M. The Inspired Homeowner Age: 38	Goals
	• Transform her tired back garden into a modern entertaining space • Get a realistic idea of cost before talking to any trades • See a visual preview so she can be confident in her decision • Find a reliable, reviewed professional without hours of research
	Pain Points
	• Has no idea how to describe what she wants in trade terms • Has been burned before by trades giving wildly inconsistent quotes • Doesn't want to commit to multiple site visits to get rough figures • Finds current platforms like Checkatrade overwhelming
Tech Level: Moderate. Comfortable with smartphones and apps; uses Instagram for design inspiration.	

Persona 2 — The Budget-Conscious Renovator

David K. The Budget-Conscious Renovator Age: 52	Goals
	• Replace a cracked driveway before winter • Stay within a strict budget and avoid bill shock • Compare multiple quotes quickly and transparently • Understand what is included in each quote
	Pain Points
	• Sceptical of online platforms — worried about being overcharged • Previous experience of vague quotes that ballooned after work started • Does not want to waste time on back-and-forth conversations • Limited time due to full-time work
Tech Level: Low-to-moderate. Uses email and basic web browsing; prefers desktop over mobile.	

Persona 3 — The Design-Led Young Professional

Priya S. The Design-Led Young Professional Age: 29	Goals
	• Create a stylish, Instagram-worthy garden or room that reflects her taste • Use technology to explore design options before committing • Book and manage everything digitally with minimal phone calls • Share the design concepts with her partner for joint decision-making
	Pain Points
	• Does not know where to start when briefing a tradesperson • Frustrated that existing platforms feel dated and text-heavy • Wants to see what the finished result could look like before spending
Tech Level: High. Early adopter; uses multiple apps daily; shares content on social media.	

4.2 Trade Professional Personas

Persona 4 — The Established Local Tradesperson

Mike T. The Established Local Tradesperson Age: 44	Goals
	• Maintain a steady pipeline of high-quality, local jobs • Reduce time spent chasing unqualified leads and no-show site visits • Showcase completed work to win more business • Reduce admin overhead — quoting, invoicing, communications
	Pain Points
	• Current lead platforms send poor-quality, vague enquiries • Has to spend unpaid time visiting sites just to provide a quote • Difficult to stand out from cheaper, unqualified competitors • Existing platforms charge high fees for low-quality leads

Tech Level: Moderate. Uses WhatsApp and basic smartphone apps; not comfortable with complex software.

Persona 5 — The Growth-Minded Trade Business Owner

Sarah L. The Growth-Minded Trade Business Owner Age: 36	Goals
	• Scale her landscaping business by accessing more qualified leads • Use AI design tools to differentiate her proposals and win premium jobs • Manage multiple team members and job pipelines efficiently • Build an online reputation that justifies premium pricing
	Pain Points
	• Hard to compete with sole traders who undercut on price • No single tool that handles leads, design, quoting, and reviews • Time-consuming to produce professional-looking quote documents
	Tech Level: High. Comfortable with digital tools; uses project management software and CRM.

4.3 Admin Persona

Persona 6 — The Platform Operator

James B. Platform Operator Age: 31	Goals
	• Ensure the marketplace runs smoothly with verified, quality trades • Monitor platform health, fraud signals, and content moderation • Configure pricing, subscription tiers, and promotional campaigns • Generate reports for business decisions and investor updates
	Pain Points
	• No clean dashboard to monitor all platform activity in one place • Difficult to spot fraudulent or low-quality trade accounts quickly • Need to manually handle disputes without a structured workflow
	Tech Level: High. Technical background; comfortable with data dashboards and admin tools.

5. User Journey Maps

5.1 Customer Journey — From Idea to Completed Job

The following flow illustrates the complete end-to-end journey for a customer using the Build-A-Job platform.

1

Discovery & Onboarding

Customer discovers Build-A-Job via app store, web search, social media, or word of mouth. Downloads the app or visits the website. Completes a lightweight sign-up (email/social login). A brief onboarding carousel explains the AI design feature.

2

Photo Capture & Upload

Customer selects a project category (driveway, garden, room, extension). Takes a photo directly in the app or uploads an existing image from their gallery. The app provides framing guidance to ensure the AI has sufficient context.

3

AI Design Generation

Platform analyses the image using computer vision to identify area boundaries, surface type, and surrounding context. The AI engine generates 3–6 design option previews (e.g., Modern Block Paving, Natural Stone, Resin Bound) displayed as side-by-side before/after comparisons. Customer can swipe through options, save favourites, and adjust style preferences.

4

Instant Estimate Range

Alongside each design option, the platform displays an indicative cost range (e.g., £7,000 – £10,500). This range is derived from image-based area estimation, selected design complexity, local material and labour cost data, and project type benchmarks. The customer can adjust the scope (e.g., adding lighting or planting) and see the estimate range update in real time.

5

Job Brief Creation & Submission

Customer selects their preferred design concept (or proceeds without selecting one). Completes a structured job brief: timing preference, budget band, location/postcode, additional notes. Reviews a summary of the brief. Submits to the platform — the system auto-matches to relevant, available trade professionals in the area.

6

Quote Review & Trade Selection

Customer receives notifications as trade professionals respond with quotes. Each quote card shows the trade profile, rating, quote amount, estimated duration, and optionally an AI-assisted alternative design the trade has generated. Customer can message trades directly within the platform to ask questions. Customer selects a preferred trade and confirms the engagement.

7

Job Completion & Review

After the job is completed, the customer is prompted to upload "after" photos. Customer leaves a star rating, written review, and optionally a before/after photo pair. The review is

published to the trade profile. The before/after pair can be featured on the platform for marketing purposes.

5.2 Trade Professional Journey — From Registration to Reputation

1

Registration & Verification

Trade professional signs up via web or app. Completes profile: company name, services offered, coverage areas, day rates, certifications (Gas Safe, NICEIC, etc.). Uploads portfolio photos and optionally adds AI-generated concept images for the types of jobs they do. Identity and credential verification is completed (manual review with email confirmation in MVP; automated in Phase 2).

2

Profile Enhancement (Optional Paid)

Trade can opt into a subscription tier (Pro or Premium) for enhanced search visibility, featured placement, and discounted lead credits. Trade can use the AI design tool to generate concept images to enrich their portfolio, even before receiving their first job through the platform.

3

Lead Notification & Qualification

When a job request matches the trade's profile (category, region, availability), a push notification is sent. The trade sees an incoming request card with: customer photos, AI design concept selected, estimate range, timing, and budget band. The trade can accept (proceed to quote), message the customer for more information, or decline the lead.

4

Quoting

Trade clicks through to the full job brief. Enters their quote amount, proposed start date, estimated duration, and payment terms. Optionally uses the in-platform AI design tool to generate an alternative or enhanced design concept to accompany their quote — a key differentiator. Submits the quote to the customer.

5

Negotiation & Confirmation

Customer may message the trade or counter-propose. Trade can revise their quote. Once agreed, both parties confirm via the platform. The job moves to "Confirmed" status. The trade receives a confirmation with job details, customer contact info, and site address.

6

Job Execution & Completion

During the job, the trade can upload progress photos (optional). Upon completion, the trade marks the job as complete in the platform. Customer is notified to review and upload after photos.

7

Review & Reputation Building

Customer leaves a rating and review. AI surfaces top-rated projects and before/after comparisons in the trade's profile. High-performing trades are badged as "Top Rated" and given enhanced algorithmic visibility. The trade's reputation score is updated in real time.

5.3 Admin Journey — Daily Operations

1

Platform Monitoring

Admin logs into the dashboard and reviews the health overview: new registrations (customers and trades), jobs posted and matched, leads purchased, and revenue for the day/week/month.

2

Trade Verification Queue

Admin reviews pending trade verification requests. Checks submitted credentials, identity documents, and portfolio. Approves, requests more information, or rejects with a reason. Verified trades receive a "Verified" badge on their profile.

3

Content Moderation

Admin reviews flagged content: reviews, photos, messages flagged by users or automated filters. Takes action: remove content, warn user, or suspend account.

4

Dispute Management

Admin accesses structured dispute cases raised by customers or trades. Reviews the job history, message thread, and evidence uploaded. Mediates and closes the dispute with an outcome record.

5

Configuration & Pricing

Admin configures subscription tier pricing, per-lead fees, promotional discount codes, and feature flags for A/B tests. Changes take effect immediately or on a scheduled date.

6

Reporting & Analytics

Admin generates reports on key metrics: platform GMV, lead conversion rates, trade retention, customer acquisition cost, NPS scores. Exports data for investor reporting or business reviews.

6. Functional Requirements

6.1 Authentication & Onboarding

6.1.1 Customer App

Feature	Description	Priority	Phase
Email Registration	Sign up with email + password. Email verification required before access.	P0 — MUST	Phase 1
Social Login	Sign in with Google and Apple. OAuth 2.0. Auto-populate profile fields.	P1 — SHOULD	Phase 1
Forgot Password	Email-based reset link. Token expires after 30 minutes.	P0 — MUST	Phase 1
Onboarding Carousel	3-slide explainer on upload, AI design, and requesting quotes. Skippable.	P1 — SHOULD	Phase 1
Profile Setup	Name, postcode, contact number. Optional profile photo.	P0 — MUST	Phase 1

6.1.2 Trade App

Feature	Description	Priority	Phase
Business Registration	Company name, contact, trading address, business type (sole trader / limited).	P0 — MUST	Phase 1
Credential Upload	Upload certifications (PDF or image). Manual review by admin in MVP.	P0 — MUST	Phase 1
Services & Coverage Setup	Multi-select service categories. Postcode-based coverage area definition.	P0 — MUST	Phase 1
Portfolio Upload	Upload up to 20 photos of completed work with labels. Before/after pairs supported.	P1 — SHOULD	Phase 1
Availability Calendar	Set available days/weeks for work. Used for smart matching.	P1 — SHOULD	Phase 1

6.2 AI Design, Visualisation & Estimation

Feature	Description	Priority	Phase
Photo Upload	Upload from camera, gallery, or file system. Accepts JPG, PNG, HEIC. Max 20MB.	P0 — MUST	Phase 1

Feature	Description	Priority	Phase
AI Image Analysis	Computer vision identifies area boundaries, surface type, and environmental context.	P0 — MUST	Phase 1
Design Option Generation	Generates 3–6 labelled design previews (e.g., Modern Oasis, Zen Retreat). Displayed as swipeable before/after cards.	P0 — MUST	Phase 1
Design Saving	Customer can save a preferred design concept to their job brief. Saved designs linked to the job throughout its lifecycle.	P0 — MUST	Phase 1
Instant Estimate Calculation	Estimate range generated per design option. Accounts for area, materials, labour benchmarks, and regional cost data. Displayed as a min-max range (e.g., £7,000 – £10,500).	P0 — MUST	Phase 1
Scope Adjusters	Optional add-ons (lighting, planting, drainage) update estimate in real time.	P1 — SHOULD	Phase 1
Trade Design Tool	Trade professionals can generate alternative AI design concepts when composing a quote, to differentiate their submission.	P1 — SHOULD	Phase 1
Before/After Showcase Generation	AI composes before/after pairs from completed jobs for platform marketing and trade profiles.	P2 — COULD	Phase 2
AR Preview	Augmented Reality overlay of design on live camera feed.	P2 — COULD	Phase 3

6.3 Job Management

6.3.1 Job Creation (Customer)

Feature	Description	Priority	Phase
Project Category Selection	Tap to select category: Driveway, Garden, Patio, Interior Renovation, Extension, Other.	P0 — MUST	Phase 1
Structured Brief Form	Fields: project description, timing preference (ASAP, 1–3 months, flexible), budget band, special notes.	P0 — MUST	Phase 1
Design Attachment	Attach one or more saved AI design options to the brief.	P0 — MUST	Phase 1

Feature	Description	Priority	Phase
Location & Postcode	Auto-detected or manually entered. Used for trade matching.	P0 — MUST	Phase 1
Job Preview & Submit	Summary screen showing all brief details before submission.	P0 — MUST	Phase 1
Job Edit (Pre-Match)	Customer can edit the brief until the first quote is received.	P1 — SHOULD	Phase 1
Job Cancellation	Customer can cancel before a trade is confirmed.	P0 — MUST	Phase 1

6.3.2 Job Matching & Lead Distribution

Feature	Description	Priority	Phase
Smart Matching Algorithm	Auto-matches job to eligible trades based on: category, coverage postcode, availability, and subscription tier. Max 5 trades notified per job in MVP.	P0 — MUST	Phase 1
Trade Search (Manual)	Customer can also browse and directly invite specific trades to quote.	P1 — SHOULD	Phase 1
Lead Credit Deduction	Per-lead fee deducted from a trade's credit balance when they view a full job brief.	P0 — MUST	Phase 1
Lead Credit Purchase	Trade can purchase lead credits via Stripe in-app. Credit packs of 5, 10, or 25 leads.	P0 — MUST	Phase 1

6.3.3 Quoting (Trade)

Feature	Description	Priority	Phase
Quote Submission Form	Fields: quote amount, VAT indication, proposed start date, estimated duration, payment terms, notes.	P0 — MUST	Phase 1
AI Design Attachment to Quote	Trade can attach an AI-generated alternative design concept to their quote.	P1 — SHOULD	Phase 1
Quote Revision	Trade can revise an unaccepted quote once per job.	P1 — SHOULD	Phase 1
Quote Withdrawal	Trade can withdraw a quote before it is accepted.	P1 — SHOULD	Phase 1

Feature	Description	Priority	Phase
Quote Comparison View (Customer)	Customer sees all received quotes side-by-side in a structured comparison layout.	P0 — MUST	Phase 1

6.3.4 Job Status Lifecycle

A job progresses through the following states:

Status	Triggered By	Description
Draft	Customer	Job brief created but not yet submitted
Open	Customer submits	Matching and notifications in progress
Quotes Received	First quote submitted	Customer reviewing quotes
Confirmed	Customer accepts quote	Trade selected; both parties notified
In Progress	Trade marks started	Work underway
Completed	Trade marks complete	Customer prompted to review
Reviewed	Customer submits review	Job closed; review published
Cancelled	Customer or Admin	Job removed from active matching
Disputed	Either party raises a dispute	Admin mediation triggered

6.4 Messaging & Notifications

Feature	Description	Priority	Phase
In-App Messaging	Real-time 1:1 messaging between customer and trade professional within the context of a job. Message history persists throughout the job lifecycle.	P0 — MUST	Phase 1
Push Notifications	Notify customers: new quote received, message received, job confirmed. Notify trades: new matched job, quote accepted, message received.	P0 — MUST	Phase 1
Email Notifications	All push notification triggers also send a fallback email.	P0 — MUST	Phase 1
SMS Notifications	Optional SMS for job confirmed and job complete events. Powered by Twilio.	P2 — COULD	Phase 2

Feature	Description	Priority	Phase
Notification Preferences	User can toggle push/email preferences per notification type.	P1 — SHOULD	Phase 1

6.5 Reviews & Reputation

Feature	Description	Priority	Phase
Star Rating	1–5 star rating. Required field before review is submitted.	P0 — MUST	Phase 1
Written Review	Optional text review. 20–500 character limit.	P0 — MUST	Phase 1
After Photo Upload	Customer uploads post-completion photos (up to 5). Before/after pairs auto-generated.	P1 — SHOULD	Phase 1
Trade Response to Review	Trade can write a single response to any review. Displayed beneath the review.	P1 — SHOULD	Phase 1
Review Moderation	Admin can remove or suppress reviews that violate platform policies.	P0 — MUST	Phase 1
AI-Highlighted Projects	AI surfaces top-rated and visually impactful before/after pairs to feature in search and profiles.	P2 — COULD	Phase 2
Top Rated Badge	Awarded to trades with 4.5+ average rating and 10+ reviews. Displayed on search results and profile.	P1 — SHOULD	Phase 1

6.6 Trade Subscription & Billing

Feature	Description	Priority	Phase
Free Tier	Trade can register and maintain a profile. No leads included. Can purchase credits individually.	P0 — MUST	Phase 1
Pro Tier (Monthly)	Includes: 10 lead credits per month, enhanced search ranking, portfolio highlighting. Recurring payment via Stripe.	P0 — MUST	Phase 1
Premium Tier (Monthly)	Includes: 25 lead credits per month, top search placement,	P1 — SHOULD	Phase 1

Feature	Description	Priority	Phase
	featured profile badge, AI design tool access.		
Credit Top-Up	On-demand credit purchases available to all tiers.	P0 — MUST	Phase 1
Billing History	Invoices and transaction history accessible in trade account settings.	P0 — MUST	Phase 1
Subscription Management	Trade can upgrade, downgrade, or cancel. Cancellation takes effect at end of billing period.	P0 — MUST	Phase 1

6.7 Search & Discovery

Feature	Description	Priority	Phase
Trade Search	Customer can search for trades by: category, postcode, availability, rating. Results paginated.	P0 — MUST	Phase 1
Trade Profile Page	Shows: name, photo, services, coverage area, rating, review count, portfolio, badges. "Invite to Quote" button.	P0 — MUST	Phase 1
Search Ranking Algorithm	Ranked by: subscription tier, rating, proximity, response rate, and profile completeness.	P0 — MUST	Phase 1
Category Browse	Browse trades by category from the home screen.	P1 — SHOULD	Phase 1
Saved Trades	Customer can bookmark a trade profile for future reference.	P2 — COULD	Phase 2

6.8 Admin Dashboard — Functional Requirements

6.8.1 Platform Overview

Feature	Description	Priority	Phase
KPI Dashboard	Real-time counters: active jobs, quotes sent today, leads purchased today, new registrations, revenue. Trend sparklines for 7/30/90 day comparisons.	P0 — MUST	Phase 1

Feature	Description	Priority	Phase
Platform Health Alerts	Automated alerts for: unusual drop in job submissions, high dispute rate, failed payment spikes.	P1 — SHOULD	Phase 1

6.8.2 User Management

Feature	Description	Priority	Phase
Customer List & Search	Search by name, email, postcode. View profile, job history, and account status.	P0 — MUST	Phase 1
Trade List & Search	Search by name, company, category, region, subscription tier. View full profile.	P0 — MUST	Phase 1
Account Actions	Admin can: suspend, reinstate, delete, or verify any user account.	P0 — MUST	Phase 1
Trade Verification Queue	List of pending verification requests with submitted documents. Approve/reject with reason.	P0 — MUST	Phase 1
Impersonate User (Read-Only)	Admin can view the platform from the perspective of any user for support purposes.	P1 — SHOULD	Phase 1

6.8.3 Job Management

Feature	Description	Priority	Phase
Job List & Search	Filter by status, category, region, date range. View full job brief and message thread.	P0 — MUST	Phase 1
Dispute Management	View open disputes. Access evidence, message thread, and job history. Close with outcome notes.	P0 — MUST	Phase 1
Manual Job Closure	Admin can force-close a job in exceptional circumstances.	P1 — SHOULD	Phase 1

6.8.4 Content Moderation

Feature	Description	Priority	Phase
Review Moderation Queue	List of flagged reviews. Preview content, approve, edit, or remove.	P0 — MUST	Phase 1

Feature	Description	Priority	Phase
Photo Moderation	Review flagged images. Remove or approve.	P0 — MUST	Phase 1
Moderation Audit Log	Record of all moderation actions with timestamp and admin ID.	P0 — MUST	Phase 1

6.8.5 Financial Management

Feature	Description	Priority	Phase
Transaction Ledger	All credit purchases, subscription payments, and refunds with full metadata.	P0 — MUST	Phase 1
Refund Processing	Admin can trigger a full or partial refund to a trade's credit balance or payment method.	P0 — MUST	Phase 1
Revenue Reports	Monthly and YTD revenue by: subscription tier, lead credits, and category. Exportable as CSV.	P0 — MUST	Phase 1
Discount Code Management	Create, edit, and retire promotional discount codes. Set usage limits and expiry dates.	P1 — SHOULD	Phase 1

6.8.6 Platform Configuration

Feature	Description	Priority	Phase
Category Management	Add, rename, archive service categories. Affects customer and trade-facing dropdowns.	P0 — MUST	Phase 1
Subscription Tier Configuration	Edit pricing, credit allocations, and feature flags per tier.	P0 — MUST	Phase 1
Estimate Parameters	Configure baseline cost data (materials, labour rates) per category and region used in AI estimation.	P0 — MUST	Phase 1
Feature Flags	Toggle specific platform features on/off by user segment or region for A/B testing.	P1 — SHOULD	Phase 1
Email Template Management	Edit transactional email content and branding via a WYSIWYG editor.	P1 — SHOULD	Phase 1

7. User Stories & Acceptance Criteria

7.1 Customer — AI Design & Estimation

User Story	Acceptance Criteria
As a customer, I want to upload a photo of my garden so that I can see AI-generated design concepts before speaking to any trades.	GIVEN I am on the Create Job screen WHEN I select "Upload Photo" or "Take Photo" THEN the app accepts the image and displays a processing indicator AND within 30 seconds, 3 or more design concept previews are shown AND each preview is labelled with a design style name AND a before/after swipe comparison is available for each concept.
As a customer, I want to see an instant cost estimate so that I know whether a project is within my budget before I commit.	GIVEN a design concept has been generated WHEN the concept card is displayed THEN an estimate range (min–max in GBP) is shown beneath each concept AND the range reflects the selected category, design complexity, and user's postcode AND a disclaimer states the range is indicative only.
As a customer, I want to save a design concept to my job brief so that the trades I contact understand my vision.	GIVEN I am viewing design concepts WHEN I tap "Save Design" THEN the selected concept is attached to my job brief AND the design image is visible to any trade who views my job request.

7.2 Customer — Job Submission & Quotes

User Story	Acceptance Criteria
As a customer, I want to submit a structured job brief so that trades have all the information they need to quote accurately.	GIVEN I have completed the job brief form WHEN I tap "Submit Job" THEN the brief is saved and the status changes to "Open" AND up to 5 matched trades are notified within 5 minutes AND I receive a confirmation notification. AND I can view the job in "My Jobs" with status "Open".
As a customer, I want to compare quotes side by side so that I can make an informed hiring decision.	GIVEN one or more quotes have been received WHEN I open "My Jobs" and select the job THEN I see a quotes comparison view AND each quote card shows: trade name, rating, quote amount, estimated duration, and a "View Profile" button AND if the trade has attached an AI design concept, a thumbnail is shown.

7.3 Trade Professional — Leads & Quoting

User Story	Acceptance Criteria
As a trade professional, I want to receive notifications for matched job requests so that I can respond quickly to relevant leads.	GIVEN a customer has submitted a job matching my profile WHEN the matching algorithm identifies my account as eligible THEN I receive a push notification and email within 5 minutes AND the notification shows the job category, postcode, and estimate range AND tapping the notification opens the job lead card.
As a trade professional, I want to submit a quote with an optional AI design attachment so that my proposal stands out from competitors.	GIVEN I am viewing a full job brief WHEN I complete the quote form and optionally attach an AI design THEN the quote is submitted and the customer is notified AND my lead credit balance is deducted by 1 AND the quote status shows as "Submitted" in My Leads.

7.4 Trade Professional — Profile & Reputation

User Story	Acceptance Criteria
As a trade professional, I want to build a rich profile with portfolio images so that customers trust my workmanship before contacting me.	GIVEN I am in Profile Settings WHEN I upload images and complete all profile sections THEN my profile completeness score reaches 100% AND my profile appears in customer search results AND portfolio images are visible on my public profile page.
As a trade professional, I want to respond to a customer review so that I can address any concerns professionally.	GIVEN a review has been posted on my profile WHEN I navigate to My Reviews and tap "Respond" THEN I can enter a response of up to 300 characters AND the response is published beneath the review AND I can only submit one response per review.

7.5 Admin — Verification & Moderation

User Story	Acceptance Criteria
As an admin, I want to review and approve trade credential submissions so that only verified professionals can accept paid leads.	GIVEN a trade has submitted credentials WHEN I open the Verification Queue THEN I can view the submitted documents and profile details AND I can Approve (grants Verified badge), Request More Info (sends email to trade), or Reject (with reason) AND the trade is notified of the outcome by email within 24 hours.
As an admin, I want to remove a flagged review so that platform content remains authentic and policy-compliant.	GIVEN a review has been flagged WHEN I open the Moderation Queue and review the content THEN I can Remove (hides the review), Approve (clears the flag), or Escalate AND all actions are logged in the moderation audit trail with timestamp and my admin ID.

8. Application Navigation & Menu Structure

8.1 Customer App — Navigation

Nav Item	Screen	Key Actions
Home	Dashboard / Feed	Start a project, browse categories, view active jobs
My Jobs	Job List	View all submitted jobs, status, and quote count
My Jobs > Job Detail	Job Detail + Quotes	View brief, compare quotes, message trades, accept quote
Design Studio	AI Upload & Preview	Upload photo, view AI designs, adjust scope, get estimate
Find a Trade	Trade Search	Search by category, postcode, rating; view profiles
Messages	Inbox	All job message threads; unread badge count
Profile	Account Settings	Personal details, notification preferences, payment methods, help

8.2 Trade Professional App — Navigation

Nav Item	Screen	Key Actions
Dashboard	Performance Overview	Leads received, quotes sent, jobs won, revenue, rating
My Leads	Lead Inbox	View incoming job requests; accept/decline/message
My Leads > Lead Detail	Full Job Brief	View photos, AI design, estimate range; compose and submit quote
My Jobs	Active & Completed Jobs	Track job status, view customer messages, mark complete
Design Studio	AI Design Tool	Generate AI design concepts for portfolio or quotes
Profile	Public Profile Editor	Edit services, coverage, rates, portfolio, certifications
Reviews	Review Management	View all ratings and reviews; respond to reviews
Billing	Credits & Subscription	Buy lead credits, manage subscription, view invoices
Messages	Inbox	Job-linked message threads
Settings	Account Settings	Notifications, password, payment methods, help & support

8.3 Admin Dashboard — Navigation

Nav Item	Screen	Key Actions
Overview	KPI Dashboard	Platform health metrics, alerts, trend charts
Users	Customer & Trade Lists	Search, view, suspend, verify accounts
Users > Verification Queue	Pending Verifications	Review and approve/reject trade credential submissions
Jobs	All Jobs List	Filter, search, view, and manage all platform jobs
Disputes	Dispute Queue	Review open disputes, take action, close with outcome
Moderation	Content Queue	Review flagged reviews, photos, and messages
Finance	Transaction Ledger & Reports	View all payments, issue refunds, generate revenue reports
Config	Platform Settings	Categories, pricing, estimate parameters, feature flags
Reports	Analytics & Exports	KPI reports, cohort analysis, CSV export
Admin Users	Team Management	Add/remove admin accounts, set permission roles

9. Non-Functional Requirements

Category	Requirement	Target / KPI
Performance	AI design generation response time	Under 30 seconds for 95% of requests
Performance	API response time (standard endpoints)	Under 500ms at p95
Performance	App page load time (cold start)	Under 3 seconds on 4G connection
Scalability	Concurrent users (Phase 1 MVP target)	Support 1,000 concurrent users without degradation
Scalability	Horizontal scaling architecture	Stateless API services deployable behind load balancer
Reliability	Platform uptime SLA	99.9% monthly uptime (excluding scheduled maintenance)
Reliability	Scheduled maintenance window	Maximum 2 hours per month; communicated 48 hours in advance
Security	Authentication standard	JWT tokens; OAuth 2.0 for social login; tokens expire after 24 hours
Security	Data encryption	All data encrypted in transit (TLS 1.3) and at rest (AES-256)
Security	PCI DSS compliance	All payment processing via Stripe; no card data stored on Build-A-Job servers
Security	GDPR compliance	Right to erasure, data portability, consent management, DPA in place
Accessibility	WCAG 2.1 Level AA	All customer and trade-facing screens pass AA accessibility audit
Mobile	iOS minimum version	iOS 15+
Mobile	Android minimum version	Android 10 (API Level 29)+
Localisation	Phase 1 locale	English (United Kingdom); GBP currency; UK date formats
Storage	User-uploaded image storage	Cloud object storage (AWS S3 or equivalent); CDN delivery
Monitoring	Error tracking	Real-time error monitoring (e.g., Sentry); alerts for error rate spikes
Monitoring	Audit logging	All admin actions logged with timestamp, user ID, and action detail

10. Technical Architecture Overview

 This section provides high-level architecture guidance. Detailed technical specifications will be produced by the engineering team based on this PRD.

10.1 Technology Stack (Recommended)

Layer	Recommended Technology	Rationale
Mobile App (iOS)	React Native / Swift	Cross-platform with native performance; aligns with Android build
Mobile App (Android)	React Native / Kotlin	Code sharing with iOS; native modules for camera access
Web App (Customer)	Next.js (React)	SEO-friendly SSR; fast initial load; component reuse
Web App (Admin)	React + Admin UI Library	Rapid dashboard development; data table components
API Layer	Node.js (Express / Fastify)	High throughput; large talent pool; async I/O for AI calls
AI / Vision Service	Python (FastAPI) + AI APIs	Python ecosystem for ML; OpenAI Vision / Stable Diffusion integration
Database	PostgreSQL (primary)	Relational integrity for jobs, users, and billing data
Object Storage	AWS S3 + CloudFront CDN	Scalable image storage; global CDN delivery for fast image loads
Payments	Stripe	PCI DSS compliant; supports UK direct debit and card payments
Push Notifications	Firebase Cloud Messaging	Cross-platform push delivery; free tier sufficient for Phase 1
Email Service	SendGrid / AWS SES	Transactional email delivery; template management
Authentication	Auth0 or custom JWT	Managed auth with social provider support

10.2 Integration Points

- AI Design Engine: OpenAI Vision API (image analysis) + image generation model for design previews. Cost estimated per-generation; batch processing for off-peak requests.
- Stripe: Subscriptions, one-time lead credit purchases, refunds. Webhooks for payment event handling.
- Google Maps / Postcode.io: Postcode validation, location-based trade matching radius calculation.
- Firebase Cloud Messaging: Cross-platform push notification delivery.
- SendGrid: Transactional email delivery for all notification types.

- Twilio (Phase 2): SMS notifications for confirmed jobs and completions.

11. Core Data Model

The following outlines the primary data entities and their key attributes. Detailed schema design will be handled by the engineering team.

Entity	Key Attributes
User	id, email, password_hash, role (customer trade admin), name, phone, postcode, profile_photo_url, created_at, status (active suspended)
Trade Profile	id, user_id, company_name, bio, services[], coverage_postcodes[], day_rate, certifications[], portfolio_images[], subscription_tier, lead_credits, verification_status, rating_avg, rating_count
Job	id, customer_id, category, status, description, timing_preference, budget_band, postcode, design_concept_id, photos[], created_at, updated_at
Design Concept	id, job_id, label, before_image_url, after_image_url, estimate_min, estimate_max, created_at
Quote	id, job_id, trade_id, amount, vat_included, start_date, duration_days, payment_terms, notes, design_concept_id, status (submitted accepted rejected withdrawn)
Message	id, job_id, sender_id, recipient_id, body, sent_at, read_at
Review	id, job_id, reviewer_id, reviewee_id, rating (1-5), body, after_photos[], trade_response, status (pending published removed), created_at
Lead Credit Transaction	id, trade_id, type (purchase deduction refund), amount, balance_after, job_id (if deduction), stripe_payment_intent_id, created_at
Subscription	id, trade_id, tier (free pro premium), stripe_subscription_id, current_period_start, current_period_end, status
Dispute	id, job_id, raised_by_id, reason, evidence_urls[], status (open resolved), outcome_notes, admin_id, created_at, resolved_at
Admin Action Log	id, admin_id, action_type, target_entity, target_id, notes, created_at

12. Security & Compliance

12.1 Data Protection & GDPR

- All personal data is processed under a lawful basis (contract performance or legitimate interest) as documented in the platform Privacy Policy.
- Users have the right to access, correct, and erase their personal data. A "Delete My Account" function must be available in profile settings.
- Data portability: customers can export their job history and reviews in JSON or CSV format.
- A Data Processing Agreement (DPA) must be in place with all third-party processors (Stripe, SendGrid, Firebase, AI providers).
- A cookie consent banner is required on the web platform before any non-essential cookies are set.

12.2 Payment Security

- No card data is ever transmitted to or stored on Build-A-Job servers. All card data is tokenised by Stripe.
- PCI DSS SAQ-A compliance applies as the platform uses hosted Stripe payment fields.
- All Stripe webhook events are verified using the Stripe webhook signature.

12.3 User-Generated Content

- All uploaded images are scanned for CSAM via a third-party API (e.g., PhotoDNA) before being stored or displayed.
- Profanity filtering is applied to all review text and messages at submission.
- Users can flag any content (review, photo, message) directly from the UI.

12.4 Platform Security

- All API endpoints require authentication except public trade profiles and the registration flow.
- Role-based access control (RBAC) enforces admin, trade, and customer permission boundaries at the API layer.
- Rate limiting is applied to all public endpoints to prevent abuse.
- Penetration testing is required before Phase 1 launch.

13. Analytics & Success Metrics

13.1 Key Performance Indicators (Phase 1)

Metric	Target (90 days post-launch)	Measurement Method
Customer Registrations	500	User table count by role
Trade Registrations	100	User table count by role
Jobs Submitted	300	Job table count by status != Draft
Quote-to-Match Rate	> 60%	Jobs with 1+ quote / total open jobs
Job Conversion Rate	> 25%	Confirmed jobs / total open jobs
Average Trade Rating	> 4.2 stars	Average of all published reviews
AI Design Engagement Rate	> 70%	Jobs with a saved design concept / total jobs
Trade Subscription Conversion	> 20% of registered trades	Trades on Pro or Premium / all trades
NPS Score (Customer)	> 40	In-app NPS survey after job completion
Platform Uptime	> 99.9%	Uptime monitoring tool (e.g., Statuspage)

13.2 Analytics Implementation

- All user events tracked via a server-side analytics pipeline (e.g., Segment) for clean, reliable data.
- Key funnel events: sign_up, photo_uploaded, design_generated, job_submitted, quote_received, job_confirmed, review_submitted.
- Admin dashboard displays real-time KPI counters and 7/30/90-day trend charts.
- A/B test instrumentation in place from Phase 1 via feature flags.

14. Phased Delivery Roadmap

Phase	Focus	Key Deliverables	Est. Duration
Phase 1 — MVP	Driveways + key exterior trades; 2–3 pilot cities	Customer app (web + mobile), Trade app (web + mobile), Admin dashboard, AI design + estimation, Job lifecycle, Stripe payments, Reviews	5–6 months
Phase 2 — Scale	Broader trade categories; national rollout	Landscaping, patios, small renovations, interiors; paid subscription tiers live; city-by-city trade onboarding; SMS notifications; automated trade verification	4–6 months
Phase 3 — Ecosystem	Advanced AI; supplier & finance integrations	AR design previews; richer room/renovation AI; supplier sponsored packs; payment/escrow; white-label option; API for partner integrations	Ongoing

15. Assumptions & Constraints

15.1 Assumptions

1. The AI design generation API (e.g., OpenAI Vision + image generation) will be commercially available and within budget for Phase 1 volumes.
2. Stripe supports all required payment flows for UK businesses including GBP subscriptions and credit purchases.
3. Trade professionals have access to a smartphone capable of running the mobile app.
4. Phase 1 will focus on the UK market only; multi-currency and multi-language support are out of scope for Phase 1.
5. Manual trade credential verification is acceptable for Phase 1. Automated verification (via third-party services) is a Phase 2 requirement.
6. The client will supply or approve brand guidelines, colour palette, and logo assets prior to UI design commencing.

15.2 Constraints

- Phase 1 budget and timeline must be validated against the feature set defined in this document during estimation.
- GDPR compliance is non-negotiable and must be addressed before any customer data is collected in production.
- The platform must not facilitate direct off-platform payments to avoid regulatory and liability issues.
- AI-generated design images must clearly indicate they are AI-generated concepts, not photographs of completed work.

16. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
AI design quality is poor or inconsistent, reducing customer trust	Medium	High	Prototype and test AI quality with real photos before committing to architecture; build a quality feedback loop
Cold-start problem: insufficient trades in a city at launch	High	High	Pre-onboard trades before public launch in each city; offer free leads as incentive during pilot period
AI cost overrun due to high volume of design generation requests	Medium	Medium	Implement a free daily generation limit for non-registered users; cache common design types
Trade fraud or fake profiles undermining customer trust	Medium	High	Manual verification for Phase 1; automated third-party identity checks in Phase 2; suspension process in place
Low customer retention after first job (no repeat usage incentive)	Medium	Medium	Focus on multi-job use cases (driveways + gardens); loyalty programme considered for Phase 2
GDPR non-compliance leading to regulatory action	Low	Critical	Legal review of Privacy Policy and cookie implementation before launch; DPAs with all processors

17. Glossary

Term	Definition
AI Design Preview	An AI-generated image showing a visual concept of a completed project, displayed as a before/after comparison.
Build-A-Job	The platform name and brand. Also referred to as BAJ internally.
Credit (Lead Credit)	A digital token purchased by a trade professional. One credit is deducted each time the trade views a full job brief.
Customer	A homeowner or property owner who uses the platform to find and hire trade professionals.
Design Concept	A specific AI-generated design option attached to a job brief, including a style name, preview images, and an estimate range.
Estimate Range	An indicative cost band (min–max in GBP) generated by the AI engine based on the job type, design complexity, and local market data.
Job Brief	The structured summary of a customer's project, including photos, AI design concept, timing, budget band, and postcode.
Lead	A job request that is sent to a trade professional for review and potential quoting.
MVP	Minimum Viable Product. The first public release of the platform with core features only.
PRD	Product Requirements Document. This document.
Quote	A formal price submission by a trade professional in response to a job brief.
Smart Matching	The algorithm that automatically identifies eligible trade professionals for a submitted job based on category, region, availability, and tier.
Subscription Tier	A monthly paid plan (Free, Pro, or Premium) that determines a trade professional's lead allocation and platform visibility.
Trade Professional	A registered business or sole trader who uses the platform to receive leads and win home improvement jobs.
Top Rated Badge	A platform badge awarded to trade professionals with 4.5+ average rating and a minimum of 10 reviews.
Verified Badge	A badge displayed on a trade profile that confirms their credentials and identity have been reviewed and approved by Build-A-Job admin.

18. Document Approval

This document requires formal review and sign-off by the following stakeholders before engineering work commences.

Name	Role	Date	Signature
Samuel Smith	Founder / Product Owner		
	Lead Product Manager		
	Lead Solutions Architect		
	Lead UI/UX Designer		
	Engineering Lead		

Version Control: All revisions to this document must be tracked with a version number, change description, author, and date in a change log maintained separately by the project team.